

12. Age - Sex Reporting Accuracy Indices

Among the basic information collected in a census, the accuracy of data on age and sex can be estimated using several indices. In data analysis, Myers' Index and Whipple's Index are commonly used for this purpose. These indices provide an understanding of the accuracy of census data.

12.1 Myers' Index

Myers' Index is a principal numerical measure used to assess the accuracy of age reporting in population data. The Myers' index measures the tendency for age to converge to single-digit values when considering absolute age, or to deviate from those values when reporting age. The Myers' index is calculated by measuring the extent to which the distribution of the last digit of a person's age, 0-9, deviates from the expected 10 percent distribution. The closer the index value is to 0, the more accurate the age data is reported, while the accuracy of the age data gradually decreases as the index value increases.

The accuracy of the data according to the value of the Myer's Index can be shown as follows.

Myers' Index Value	Level of Accuracy
Less than 6	Data with a high degree of accuracy (The data are very slightly affected by heaping/ There is less tendency to assemble towards units digit to deviate from such numbers.)
6 - 20	Data with a less degree of accuracy (Data are under the influence of heaping/ tends to assemble towards units digit or to deviate from such numbers.)
Greater than 20	Data with poor accuracy (Data are obviously affected by heaping/ greater tendency to assemble towards units digit or to deviate from such numbers.)

Table 12.1 : Myer's Index by Sex, 1981, 2001, 2012 and 2024

Sex	Myers' Index			
	1981	2001*	2012	2024
Both sexes	9.7	2.7	1.7	1.2
Male	8.7	2.7	1.8	1.2
Female	11.2	3.0	1.7	1.1

*Based on 18 districts only.

According to Table 12.1, the Myers' Index for the total population in 2024 is 1.2. The index value is 1.1 for females and 1.2 for males. Compared with the 2012 Census, the index has further declined in 2024 for both sexes. This decline indicates an improvement in the accuracy of age reporting in the 2024 Census. The very low index values suggest that only a minimal proportion of the population exhibited digit preference or age misstatement. Accordingly, the age and sex data collected in the 2024 Population and Housing Census can be considered highly reliable based on Myers' Index.